Computational Intelligence in Engineering

Week 09 – Project B: Introduction

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CONTENT

Overview

- Key Takeaways from Project A
- Project B: Introduction
  - Real Life example
  - Motivation
  - Problem Statement
- Plan of action

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Project A: Overview

Key Take aways from Project A:

- Dealing with sequential data
- Data manipulation and preprocessing
- Choosing relevant features
- Performing classification using various NN architectures
- Calculating performance metrics of the model
- Manual tuning of hyperparameters and comparing the model performance

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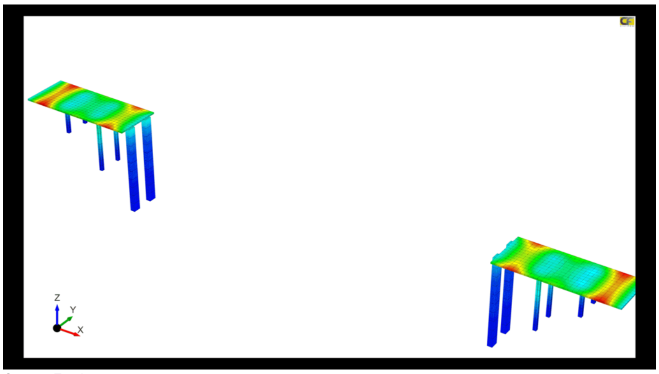
Real Life example: Hohenzollernbrücke

*Source: Wkipedia

*Source: Köln tourism

Motivation

- Get dynamic and realtime information on the structural health
- Predicting future failures and performing preventive maintenance
- Identifying premature failures
- Efficient manpower and task scheduling to focus on probable failure zones rather than complete inspection of the structure



Source: Enteknograte

Monitoring and tracking of a suspension railway based on inertial measurements and computational intelligence

R&D @ IAM

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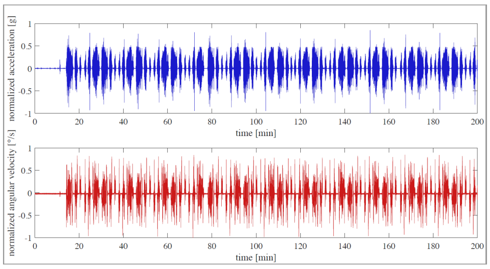
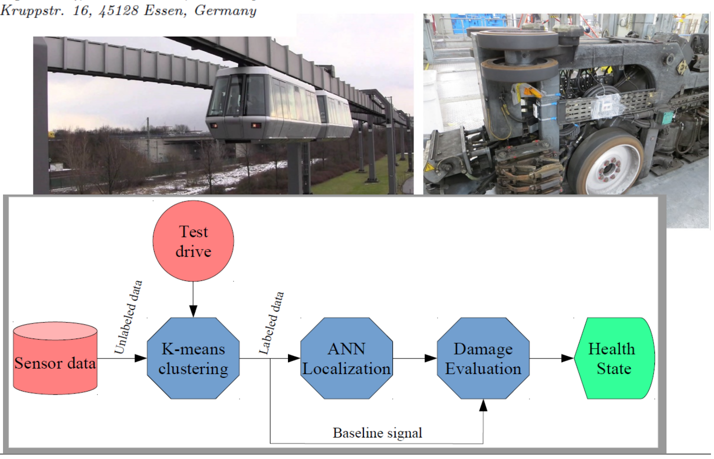




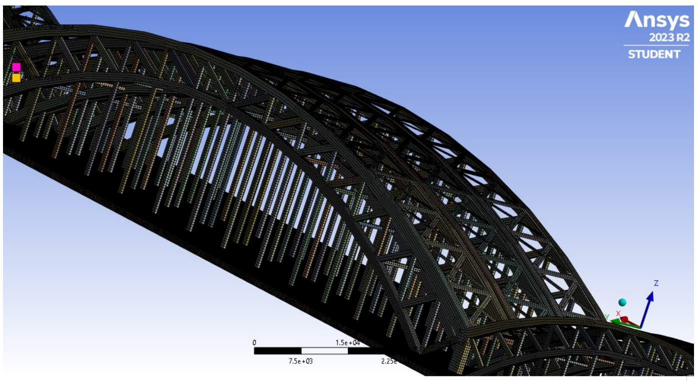
Problem Statement

Given

- Equivalent 3D model of Hohenzollernbrücke
- Gravitational force and self weight of the bridge is considered
- Loads contributed by the trains
- Average load due to pedestrians
- Load due to crosswinds acting on the bridge.
- Simulation data for perfect & imperfect bridge structure for various loads, magnitudes of crosswind and average pedestrian loads

Task

- Determine if the structure has some imperfections and locate their exact position



Data

Node Number	Directional Deformation (mm) for X_axis	Directional Deformation (mm) for X_axis 0.1	Directional Deformation (mm) for X_axis 0.5	Directional Deformation (mm) for X_axis 0.9	Directional Deformation (mm) for X_axis 1.1	Directional Deformation (mm) for X_axis 1.5	Directional Deformation (mm) for X_axis 1.9	Directional Deformation (mm) for X_axis 2.1
1	2.14E-07	-7.99E-04	-3.46E-04	-7.97E-04	-2.98E-05	-3.82E-05	-2.97E-05	3.51E-04
2	2.92E-07	-5.36E-04	2.22E-04	-5.34E-04	-7.89E-06	5.81E-06	-7.76E-06	2.70E-04
3	4.83E-07	-3.60E-05	1.28E-03	-3.37E-05	1.63E-05	5.43E-05	1.65E-05	1.54E-04
4	2.60E-07	2.51E-05	1.45E-03	2.74E-05	1.76E-06	2.51E-05	1.89E-06	-7.73E-06

Given

- The zip folder contains the data files for Perfect and Imperfect structure
- The magnitude of the load applied is given on the name of each folder along with the loading condition (Eg: The folder with name Imperfect_no_1_1_train_extreme_track_2000N contains simulation results for bridge with imperfection on location 1 and 1 train travelling on the extreme outside track and applying a load of 2000N)
- The location of the Imperfection can be seen in the figures provided (the exact nodal location can be calculated from the data provided)
- Each folder has data for deformation, shear stress in X,Y and Z direction as well as total deforamtion
- A sample excel sheet is shown in figure above. It contains the information for node number and respective deformation in X-direction at 0.1 s,0.5 s,0.9 s….. timesteps

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Plan of action

- Data pre-processing and selecting relevant features
- Baseline AI/ML model
- Validation set to check for overfitting
- Hyperparameter tuning
- Cross validation to determine the sensitivity of the model
- Calculating performance metrics (e.g. R2 score in case of regression and Confusion matrix, ROC curves, AUC in case of classification)

Thanks for your attention!